Torrey Zaches

poppyzaches@gmail.com | linkedin.com/in/torrey-zaches | github.com/Torrey0

EDUCATION

California Polytechnic State University, San Luis Obispo San Luis Obispo, CA | Class of 2027

B.S. Computer Engineering, B.S. Mathematics, GPA: 3.94

Relevant Coursework: Graduate Digital Design, Computer Architecture, GPU Architecture, Hardware Security Embedded ML, Operating Systems, Computer Networks, Database Systems, Advanced Linear Algebra, PDEs

SKILLS

- C, C++, FreeRTOS, Rust, Embassy, RISC-V, SystemVerilog, Java, Python, SQL, R, Autodesk Maya

EXPERIENCE

CAL POLY PROTOTYPE VEHICLE LABORATORY San Luis Obispo, CA | Oct 2023 – Present

- Leading the Low-Voltage Team to create a long-distance electric sports car
- Designing schematics for car sensors and telemetry (IMU, humidity, pedals, SDIO logging) and writing corresponding ESP-IDF firmware to collect data using continuous ADC, I2C, and SPI
- Developing parameterizable code for a network of ESP-IDF and Embassy nodes to schedule sensor data collection, monitor one another via CAN bus, and disconnect high-voltage power in emergencies
- Engineering an end-to-end telemetry pipeline by implementing a custom LoRa data streaming protocol and developing a Java Swing dashboard to display real-time sensor data and send commands remotely
- Specced LiFePO4 cells and secure storage, and wired a REC BMS to safely charge and monitor over CAN
- Configured an OpenWrt router to drive a USB MIMO transceiver, bringing Wi-Fi to PROVE's garage

ISA - OPERATING SYSTEMS / MICROCONTROLLERS San Luis Obispo, CA | Apr 2025 – Present

- Leading PLTL workshops, guiding students through exercises, and promoting effective problem-solving
- Strengthening technical communication skills by explaining key concepts and helping students debug

CHARTWELLS (CAL POLY CAMPUS DINING) San Luis Obispo, CA | Sep 2023 – May 2024

- Prepared orders and assisted customers in a fast-paced setting while maintaining high service standards

PROJECTS

BCRYPT HARDWARE ACCELERATOR SystemVerilog

- Designed a Basys-3 FPGA accelerator computing 145 work-factor-10 BCrypt hashes per second
- Received target passwords over UART and compared with a dictionary stored on an SPI Flash

TINYGPU ARCHITECTURE EXTENSION SystemVerilog

- Created a hardware round-robin priority scheduler to enable concurrent execution of multiple kernels
- Implemented a hardware reconvergence stack to add support for nested divergence
- Developed a Cocotb Python host controller to use and test the design

RISC-V MICROPROCESSOR SystemVerilog

- Engineered a five-stage pipelined RISC-V 32I microprocessor
- Devised a set-associative cache for data memory and a direct-mapped cache for instruction memory

PATH FOLLOWING AUTONOMOUS CAR Embedded C

- Built a compact 6” × 6” car that utilized a PID controller to follow a path of a target color
- Programmed an STM32 microcontroller to receive data from an array of light sensors over I2C

3D MODELING IN AUTODESK MAYA Autodesk Maya

- Created and animated a detailed model of the Octane car from Rocket League

HONORS & AWARDS

- Edward Van Duyne Scholarship (2025, 2026)
- Eagle Scout, Boy Scouts of America